

The Cleavage Strength of Polycrystals

By

N. J. Petch, B.Sc., B.Met., Ph.D.

SYNOPSIS

Using mild steel, ingot iron, and spectrographic iron, it is found that the cleavage strength σ_c is related to the grain size l by

$$\sigma_c = \sigma_0 + k l^{-1/2}$$

This can be explained if the Griffith cracks can be identified with glide planes on which dislocation movement has been blocked by the grain boundaries.

According to this theory, the cleavage strength of the polycrystals is controlled by the theoretical cleavage strength in the grain-boundary region, which opens up the possibility of subtle composition effects in this special region, by the grain size, which determines the stress-concentration factor, and by σ_0 . The plastic strain at fracture is greater when the grains are finer and the strain-hardening rate is lower.

744

It is well known that coarse-grained steels are more prone to brittle fracture than fine-grained ones, and the present paper describes a detailed examination of this feature, undertaken in the hope that an understanding of it might throw a light on the whole subject of brittle fracture.

Notched specimens of various forms are often used in this type of investigation, but the occurrence of brittle fracture in such cases depends upon both the yield stress and the brittle, or cleavage, strength.¹ To separate these two properties, the tensile fracture of unnotched specimens in liquid nitrogen was used in the present work, and the cleavage strength was obtained as the fracture stress. The variation of this cleavage strength with ferritic grain size was studied.

EXPERIMENTAL DETAILS

Mild steel, ingot iron, and spectrographic iron were used, and a few measurements were also made on the mild steel after decarburization. The compositions of the metals were as follows:

	Ingot Iron	Spectrographic Iron	Mild Steel
C, %	0.036	0.07	0.155
Si, %	<0.005	<0.005	0.10
Mn, %	0.10	0.005	0.59
P, %	0.007	0.005	0.028
S, %	0.020	0.006	0.033
Ni, %	0.07	0.01	0.18
Cr, %	...	...	0.11
Mo, %	...	...	0.025
O, %	0.042	0.012	0.018
N, %	0.003	0.008	0.004

The specimens (1/4-in. dia. ends) were prepared with care; all machining marks were removed from the gauge length and the necks by polishing with 000 emery paper. The machining stresses were relieved by the subsequent heat treatment *in vacuo* that was used to establish the required grain sizes, which ranged from single crystals to 8000 grains/sq. mm. Almost all these sizes were obtained by heat treatment at temperatures up to 1050° C. for times up to 24 hr., followed by cooling at various rates. For the fastest rate, the specimens were sealed in evacuated thin-walled silica tubes, and were air-cooled. Results were also obtained from other specimens treated at temperatures up to 1350° C., but, with one exception, the fractures were found to be intercrystalline instead of cleavage, so that these fracture stresses had to be discarded. The single crystals and a few coarse-grained specimens were obtained by straining and annealing the decarburized mild steel.

To promote true axial loading during the mechanical testing, the specimens were screwed directly into polished 1/2-in. dia. ball-bearings resting in polished,

hardened tool-steel seatings. A testing machine was used in which the load was measured by the longitudinal elastic deformation of a bar, and this deformation was followed by a mirror extensometer that was illuminated by a beam of light. By this means, a direct photographic record of the stress/strain curve could be obtained.

To obtain satisfactory reproducibility, it was found that 6-12 separate counts of grain size were desirable on each specimen. Since the specimen surface was a random section through the grains, the figure derived for the grain diameter was in fact proportional to the true diameter.

RESULTS

Various amounts of plastic deformation before fracture were observed, depending upon the grain size, and some allowance for this was necessary, since the cleavage strength is altered by plastic strain. To make the results directly comparable at the various grain sizes, a correction to zero deformation was carried out using the pre-strain/cleavage-strength curves determined by McAdam, Geil, and Mebs² for an ingot iron and a low-carbon steel. These were obtained by deformation at room temperature, followed by fracturing at -188° C. This correction is not perfect; low-temperature strain produces more strain hardening than the same amount of room-temperature strain, so that there is probably an underestimate. However, the correction was negligible for about half the results and was small for many of the others, although it did reach 25 tons/sq. in. for the few specimens with a reduction in area of about 35%.

Some allowance had also to be made for the non-uniform stress distribution in the cross section when there was necking. For the few cases that occurred, the greatest stress was calculated using Bridgman's

Manuscript received 19th December, 1952.

Dr. Petch is Reader in Metallurgy in the University of Leeds.

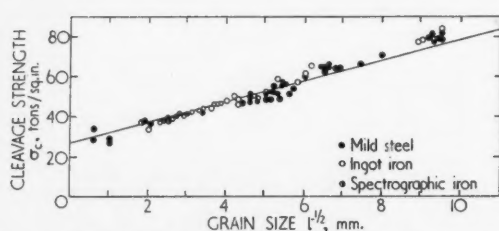


Fig. 1—Relationship between cleavage strength and grain diameter

analysis.³ The corrections were small and were in the opposite direction to the plastic-strain correction.

The cleavage-strength/grain-size results are shown in Fig. 1. Because of overlapping, this diagram does not include all the experimental points. There is a large change in cleavage strength (25–85 tons/sq. in.) over the range of grain sizes examined, and the relationship between the cleavage strength σ_c and the grain diameter l corresponds closely to the form

$$\sigma_c = \sigma_0 + kl^{-1/2}$$

where σ_0 and k are constants.

In drawing the line shown in Fig. 1, greatest weight was given to the results up to $l^{-1/2} = 5$, because the plastic-strain corrections were zero or very small. At larger $l^{-1/2}$ values (i.e., smaller grain sizes), the cleavage strengths lie a little above this line, but this is consistent with the greater plastic strains involved in these specimens and the probable under-estimate of the strain correction.

THEORY OF THE CLEAVAGE-STRENGTH/GRAIN-SIZE RELATIONSHIP

The basic problem in fracture is to understand the physical nature of the imperfections (the Griffith cracks⁴) that account for the difference between the theoretical and the observed strengths.

Actual cracks of the Griffith form in a metal would close up in the absence of external stress unless they were prevented from doing so by included atoms, so it is possible that the Griffith cracks can be identified with precipitates in the form of very flat plates. Cracks could then develop under stress owing to a low adhesion between the precipitate and matrix and to differences in their elastic and plastic properties.

There are probably cases, particularly of inter-crystalline fracture, in which precipitates form the effective Griffith cracks, and the author was first attracted to the possibility that this might be the case with the present cleavage fractures. The Griffith expression for the cleavage strength ($\sigma_c = \sqrt{(2ES/\pi c)}$, where E , S , and $2c$ are the Young's Modulus, surface energy, and crack size respectively) would then require a specific relationship between precipitate size and grain size to account for the observed variation of cleavage strength. However, a detailed examination of the cleavage strength of recrystallized zinc by Zein and the present author⁵ also gives $\sigma_c = \sigma_0 + kl^{-1/2}$ (σ_0 is very small), and it would be surprising if the same form of precipitate-size/grain-size relationship existed in both cases. Some more general theory would be more acceptable.

Such a general theory can be supplied if the Griffith

cracks can be identified with glide planes on which dislocation movement has been held up by blockage at a grain boundary. The tensile stresses at the head of such an array of dislocations can be very high, and the theoretical cleavage strength may be attained and fracture initiated. It should be possible to calculate the cleavage-strength/grain-size relationship that would result from this mechanism.

Zener,⁶ who first dealt with slip planes as stress concentrators, has compared the concentration due to relaxation of the shear stress on a slip plane with that due to a crack. In a recent paper, Eshelby, Frank, and Nabarro⁷ have calculated the positions taken up by the dislocations in an array of like dislocations that is pressed against an obstacle and the shear stresses that result. Koehler⁸ has used these dislocation positions to calculate the tensile stresses. In the present instance, this calculation will be extended to establish the cleavage-strength/grain-size relationship.

Koehler examined an array of n like dislocations pressed against an obstacle by a shear stress τ . Summation of the stresses due to the individual dislocations showed that high tensile stresses are produced in the region at the head of the array that are of the form

$$\sigma = \alpha n \tau \dots \dots \dots (1)$$

where σ is the stress and α is a constant.

To extend this result into a cleavage-strength/grain-size relationship, the connection between n and the grain size is required.

The dislocations initially present within the crystal can probably begin to move at small stresses, lower than the conventional yield point, but they cannot pass beyond the confines of the grain boundary. On the glide planes that contain a Frank-Read⁹ source, there will be dislocation multiplication, and this will continue until the accumulation of the dislocations on the glide plane and the interaction between the dislocations within the accumulation stop the multiplication for that particular applied stress. There will then be an array of n positive dislocations pressing against the grain boundary at one end of the glide plane and n negative dislocations pressing against it at the other end of the glide plane; the largest value of n will arise in grains with a source at the centre. Since the interaction between the positive and negative dislocations will be confined mainly to the centre of the grain, the two parts of the array can be treated separately, to a first approximation. The maximum value of n under any stress τ can then be calculated from the condition that the length of an array of

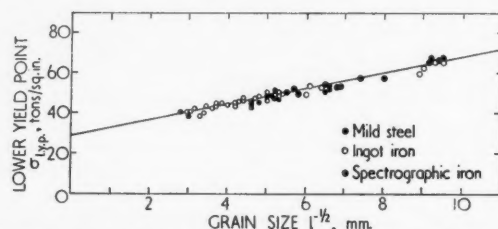


Fig. 2—Relationship between lower yield point and grain diameter

n like dislocations under this stress must be equal to half the grain diameter.

The length L of an array of like dislocations as calculated by Eshelby, Frank, and Nabarro is given by $L = 2nA/\tau$, where $A = Gb/2\pi(1 - \nu)$ and G , b , and ν are the rigidity modulus, Burgers vector, and Poisson's ratio respectively.

When there is dislocation saturation for τ , L can be identified with the semi-diameter $l/2$ of the grain.*

$$\text{Hence, } l = \frac{4nA}{\tau} \text{ and } n = \frac{l\tau}{4A}.$$

Substituting this value of n in equation (1) gives

$$\sigma = \frac{\alpha l \tau^2}{4A} \dots \dots \dots (2)$$

In the array that has been considered so far, it has been assumed that the dislocations are free to move with the applied shear stress, within the limitation of their own interaction. In real crystals, however, there will also be an internal stress field due to dislocations outside the array. Since the glide planes of interest as stress concentrators have the most densely packed dislocations, the dislocation spacing on the neighbouring planes must be wider. These neighbours will consequently introduce a stress tending to open up the densest plane to their own dislocation spacing and this stress will oppose the applied shear stress. The dislocation spacing normal to the glide plane will not be affected by the operation of the Frank-Read source, and increase in the value of τ will decrease the spacing on both the densest planes and their neighbours, so a reasonable approximation is that the internal stress can be represented by an average shear stress τ_0 in opposition to τ and independent of it.

Then, from equation (2),

$$\sigma = \frac{\alpha}{4A} (\tau - \tau_0)^2 l.$$

In terms of the tensile stresses σ^* and σ_0 corresponding to τ and τ_0 ,

$$\sigma \propto (\sigma^* - \sigma_0)^2 l.$$

When σ reaches the theoretical cleavage strength $\sigma_{t.c.s.}$,

$$\sigma^* = \sigma_c.$$

$$\sigma_{t.c.s.} \propto (\sigma_c - \sigma_0)^2 l.$$

If this is rearranged, with $\sigma_{t.c.s.}$ constant,

$$\sigma_c = \sigma_0 + k l^{-1/2} \dots \dots \dots (3)$$

where k is a constant.

Thus, provided that the theoretical cleavage strength at the grain boundary is not appreciably affected by dislocation multiplication below the yield point, equation (3) gives the cleavage-strength/grain-size relationship for zero plastic deformation. This agrees with the observed relationship.

THE YIELD POINTS

The dependence of the lower yield point on the grain size in these experiments (see Fig. 2) was of the form $\sigma_{l.y.p.} = \sigma_0 + k' l^{-1/2}$, and the results for the different materials lay on the same line. This is in

*After yielding, arrays of dislocations of like sign can probably be produced that extend right across a grain. This does not alter equation (3).

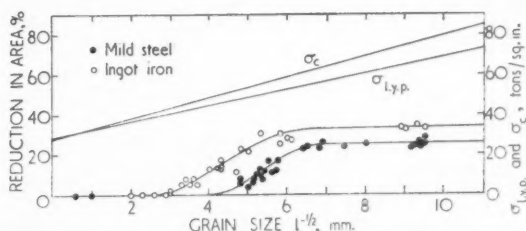


Fig. 3—Experimental curves for the dependence of the cleavage strength, lower yield point, and reduction in area on the grain diameter

agreement with Hall's room-temperature measurements.¹⁰ The upper yield point lay above the lower for fine grains, but they merged at about $l^{-1/2} = 5$.

The cleavage strength and lower yield point are plotted together in Fig. 3 without the experimental points. It will be seen that σ_0 is common.

If the ideas of the previous section are correct, this similarity between the cleavage-strength/grain-size and the yield-point/grain-size relationships is to be expected. The lower yield point, which represents the propagation of a Lüder's line, and probably also the upper yield point^{10,11} are, like cleavage, concerned with the stress concentration produced by a blocked glide plane. The yield point involves the shear stresses, whereas cleavage involves the normal stresses.

INFLUENCE OF CARBON CONTENT

According to the present theory, the cleavage strength of low-carbon steels, in which ferrite predominates in the microstructure, is determined by the fracture of one ferrite grain under the influence of dislocations in another ferrite grain. Ferrite alone is involved, and the cleavage strength, corrected for plastic deformation, should be nearly independent of the pearlite content. This is borne out by Fig. 1. The independence of carbon content shown by the yield point (see Fig. 2) is of essentially similar origin.

The presence of pearlite, however, increases the strain-hardening rate, and so it should reduce the amount of plastic strain required to raise the stress from the yield point to the cleavage strength. In agreement with this, the reduction in area (see Fig. 3) was found to be greater for the ingot iron than for the mild steel.

As would be expected, Fig. 3 also shows that the reduction in area increases for the finer grain sizes, owing to the increasing separation of σ_c and $\sigma_{l.y.p.}$. The reduction in area does not increase indefinitely, but becomes constant. The separation of σ_c and $\sigma_{l.y.p.}$, which refer to zero plastic strain, does not fully indicate what the reduction in area will be; the rates of increase of cleavage strength and of yield stress with strain and the influence of grain size on these rates are also involved.

CONCLUSIONS

The present work indicates that the cleavage strength of polycrystals is controlled by the theoretical cleavage strength in the vicinity of the grain boundaries, by σ_0 , and by the grain size. The latter determines the magnitude of the stress-concentration

factor. Since a property of the grain-boundary region is involved, subtle effects of composition are possible owing to the segregation of solute atoms to this special region and to the effect of other solute atoms on this segregation.

The amount of plastic strain at fracture, which depends partly on the separation of the yield point and the cleavage strength, will increase with finer

grain sizes (eventually becoming constant) and lower strain-hardening rates. Composition effects at the grain boundary that affect the cleavage strength may also affect the yield point and the plastic strain at fracture.

These conclusions apply so long as there is no complication arising from the intrusion of a precipitate that becomes the controlling stress concentrator.

References

1. E. OROWAN: "Reports on Progress in Physics," vol. 12, pp. 185-232: 1948-1949, London, The Physical Society.
2. D. J. MCADAM, jun., G. W. GEIL, and R. W. MEBS: *Trans. Amer. Inst. Min. Met. Eng.*, 1947, vol. 172, pp. 323-362.
3. P. W. BRIDGMAN: *Trans. Amer. Soc. Metals*, 1944, vol. 32, pp. 553-572.
4. A. A. GRIFFITH: *Phil. Trans. Roy. Soc., Ser. A*, 1920, vol. 221, pp. 163-198.
5. N. J. PETCH and F. ZEIN: Unpublished work.
6. C. M. ZENER: "Elasticity and Anelasticity of Metals," pp. 126-163: 1948, Chicago, Chicago University Press.
7. J. D. ESHELBY, F. C. FRANK, and F. R. N. NABARRO: *Phil. Mag.*, 1951, vol. 42, pp. 351-364.
8. J. S. KOEHLER: *Phys. Rev.*, 1952, vol. 85, p. 480.
9. F. C. FRANK and W. T. READ: "Symposium on the Plastic Deformation of Crystalline Solids," pp. 44-46: 1950, Pittsburgh, Mellon Institute.
10. E. O. HALL: *Proc. Phys. Soc., Ser. B*, 1951, vol. 64, pp. 747-753.
11. A. H. COTTRELL: "Symposium on the Plastic Deformation of Crystalline Solids," pp. 60-72: 1950, Pittsburgh, Mellon Institute.

Gravimetric Determination of Sulphur in Basic Steelmaking Slags

By the
Methods of Analysis Committee

SYNOPSIS

A gravimetric method for the determination of sulphur in slag, including a perchloric acid fuming technique, is recommended; baking in *aqua regia* solution has also been shown to provide acceptable results. In the presence of barium, treatment of the insoluble residue is necessary to avoid error. 743

A gravimetric method on similar principles was examined by the other group, but results on one particular slag (containing fluorine and barium sulphate) were not satisfactory, and it was found that, in the presence of barium salts, pre-precipitation of barium sulphate could occur. If this happens before or during filtration of the silica residue, low results will be obtained unless the sulphur is recovered. This premature separation of sulphate by barium ions initially present in the solution appeared to be fortuitous, depending upon a critical balance of acidity, temperature, time, and concentration. The earlier method of the combustion group apparently gave correct results because this balance of critical features was slightly more favourable.

Another method that was put forward was based on perchloric acid dehydration. With this procedure, fluorine is expelled more efficiently, and barium compounds are brought into solution by a fuming treatment. It was realized that even with perchloric acid there was a possibility of premature precipitation of barium sulphate, and that any accurate method must involve examination of the residue.

It was also suggested that an alkali fusion procedure,

IN 1949, the Sulphur Working Party of the Steel Practice Committee of the Steelmaking Division of the British Iron and Steel Research Association issued a report which showed that sulphur determination by routine methods in different laboratories produced figures which varied so widely that subsequent calculation of sulphur partition coefficients was invalidated.

The project was therefore referred to the Blast Furnace Raw Materials and Slags, Steelmaking Slags, and Refractories Analysis Sub-Committee for the development of a rapid and reliable method for the determination of sulphur in slag. Although, for control purposes in melting-shop laboratories, this was the main requirement, it was necessary first to produce a reliable standard method as a basis of comparison.

The constitution of the Sub-Committee is given in Appendix II.

PRELIMINARY WORK

To expedite the completion of the work, two groups were appointed to consider the different aspects of the problem. The combustion technique was chosen for development as a rapid method; the group examining this temporarily used, as a basis of comparison, a gravimetric method involving solution in *aqua regia* in the presence of potassium chlorate and bromine, dehydration of the silica residue by baking, re-solution in hydrochloric acid, evaporation to low bulk (10 ml.), dilution, filtration, and precipitation with barium chloride. Satisfactory results were obtained on five basic slags with sulphur contents between 0.18% and 0.51%; a detailed report on this research will be published later.

Paper MG/DA/146/52 of the Methods of Analysis Committee of the Metallurgy (General) Division of the British Iron and Steel Research Association, received 20th November, 1952.

Table I
SULPHUR CONTENT OF A BARIUM-CONTAINING
SLAG, %

Method	A	B	C	D
Analyst				
1	0.39	0.41	0.35	0.46
2	0.40	0.41	0.39	0.41
3	0.37	0.39	0.39	0.40
4	0.39	0.39	0.39	...
5	0.395	...	0.40	...
6	0.355	0.38	0.365	0.39
7	0.39	0.395	0.395	0.455
8	0.39	0.41	0.40	0.27

depending on the decomposition of the slag by sodium carbonate fusion at 1000° C. in a platinum crucible, followed by water extraction, filtration, acidification, and precipitation with barium chloride, would give satisfactory results.

COMPARISON OF METHODS

Since some members were able to obtain reasonably concordant results using either of the methods, a further sample of barium-containing slag was circulated. Sulphur was determined using four variations of the gravimetric procedure to assess their respective value (see Table I). Method A involves perchloric acid fuming; method B uses *aqua regia* decomposition and baking, and also includes evaporation after filtration of the siliceous residue; method C is similar to B, but omits evaporation; and method D is an alkali fusion procedure.

With method B, involving the evaporation after separation of silica, a precipitate or turbidity was often produced, which proved to be a hydrolysis product consisting of titanium, aluminium, iron, and some sulphur. This was avoided in method C, which omits the evaporation, although filtration had to be completed quickly.

In the absence of an oxidant, the alkali fusion method yielded unsatisfactory figures (the low result reported for analyst No. 8 was shown to be due to the incomplete conversion of sulphide to sulphate), but it was thought advisable to examine it further.

Table II
SULPHUR CONTENTS OF SLAGS, %

Method	MGS 140			MGS 147		
	A	B	D	A	B	D
Analyst						
1	0.395	0.39	0.39	0.155	0.16	0.185
2	0.37	0.385	0.39	0.13	0.135	0.135
3	0.385	0.38	0.40	0.155	0.155	0.165
4	...	...	0.40	0.165	0.165	0.185
5	0.40	0.395	0.385	0.165	0.165	0.165
6	0.39	0.39	...	...	...	...
7	0.38	0.38	0.38	0.13	0.14	0.17
8	0.385	0.38	0.41	0.16	0.16	0.17
9	0.39	0.41	0.46	...	...	...
10	0.40	0.395	0.37	0.16	0.155	0.16
11	0.41	0.40	0.395	...	...	...
12	0.365	0.39	0.40	0.15	0.15	0.155

Table III
RECOVERY OF BARIUM SULPHATE FROM SLAGS

Method	Barium Sulphate, mg.			
	Present		Recovered	
	Sulphur Equivalent in Slag	Added	Total* (less blank)	Added
Perchloric acid (Blank 1.6 mg.)	51.4	14.6	66.4	15.0
	51.4	29.2	80.8	29.4
<i>Aqua regia</i> + baking (Blank 2.4 mg.)	51.4	14.6	65.2	14.4
	51.4	29.2	81.4	30.0
Perchloric acid	55.5	...	s. 55.8	...
	55.5	48.0	s. 60.0 r. 43.4	...
			103.4†	47.6
<i>Aqua regia</i> + baking	55.5	...	s. 55.2	...
	55.5	48.0	s. 51.6 r. 51.4	...
			103.0†	47.8
Alkali fusion	27.8	...	28.6	...
	27.8	48.0	77.0	48.4
Barium recovery as barium sulphate from above samples			3.4 51.8	...
				48.4

* s. = solution, r. = residue

† Barium recovery as BaSO₄ = 42.0 and 53.8 mg. respectively

Two more slags were therefore treated by methods A (perchloric), B (*aqua regia* + baking), and D (alkali fusion with oxidant). The results are given in Table II.

A high blank was often obtained on the fusion method, and it was felt that solution methods A and B were more convenient in operation, and were capable of giving satisfactory results.

SULPHUR IN THE PRESENCE OF BARIUM

Since some basic slags contain barium, derived from fluorspar impurities, consideration was given to the complete recovery of sulphur in such slags. A residue treatment was therefore introduced into the method, and tests were carried out using slags of known sulphur content to which further barium sulphate was added, in amounts much higher than those likely to be experienced in practice.

Both the perchloric fuming and the *aqua regia* baking techniques were employed, and, as a further check, a fusion method described by Lundell, Hoffmann, and Bright* was used.

The results obtained are summarized in Table III.

* G. E. F. Lundell, J. I. Hoffmann, and H. A. Bright, "The Chemical Analysis of Iron and Steel," p. 570: 1946, New York, John Wiley and Sons.